



Aurélien Geron

P. 181

Machine Learning

Lecture 3: Support Vector Machine

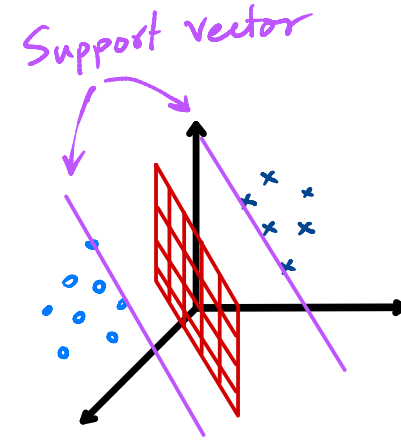
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Topics

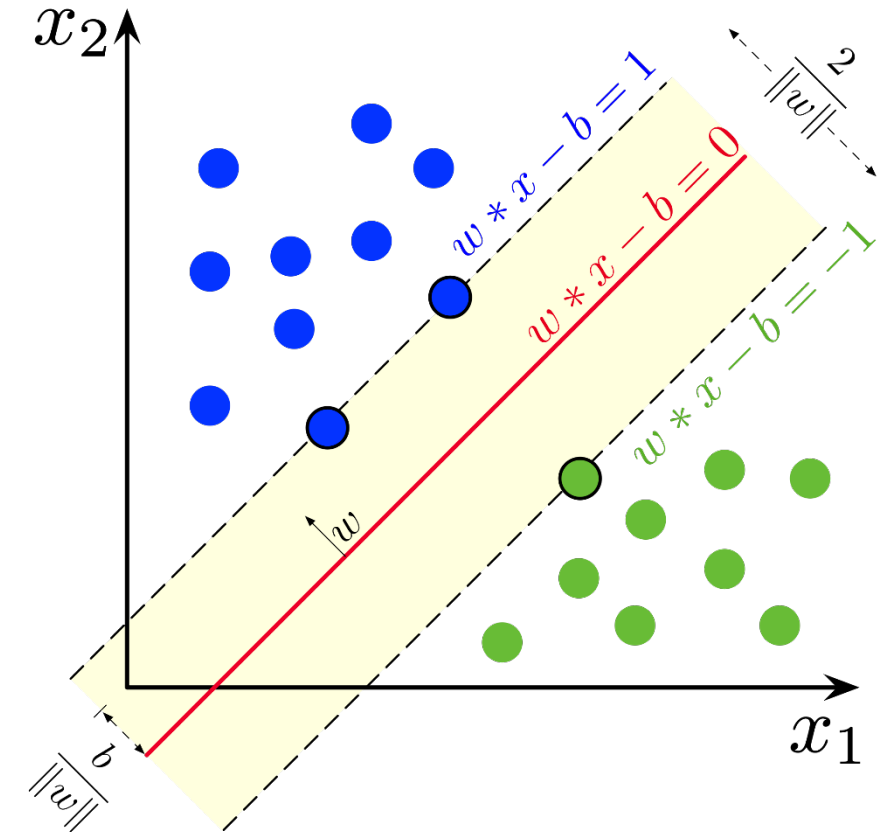
- Linear SVM classification
- Soft margin
- Training SVM
- Nonlinear SVM classification
- Kernel trick
- Hinge Loss



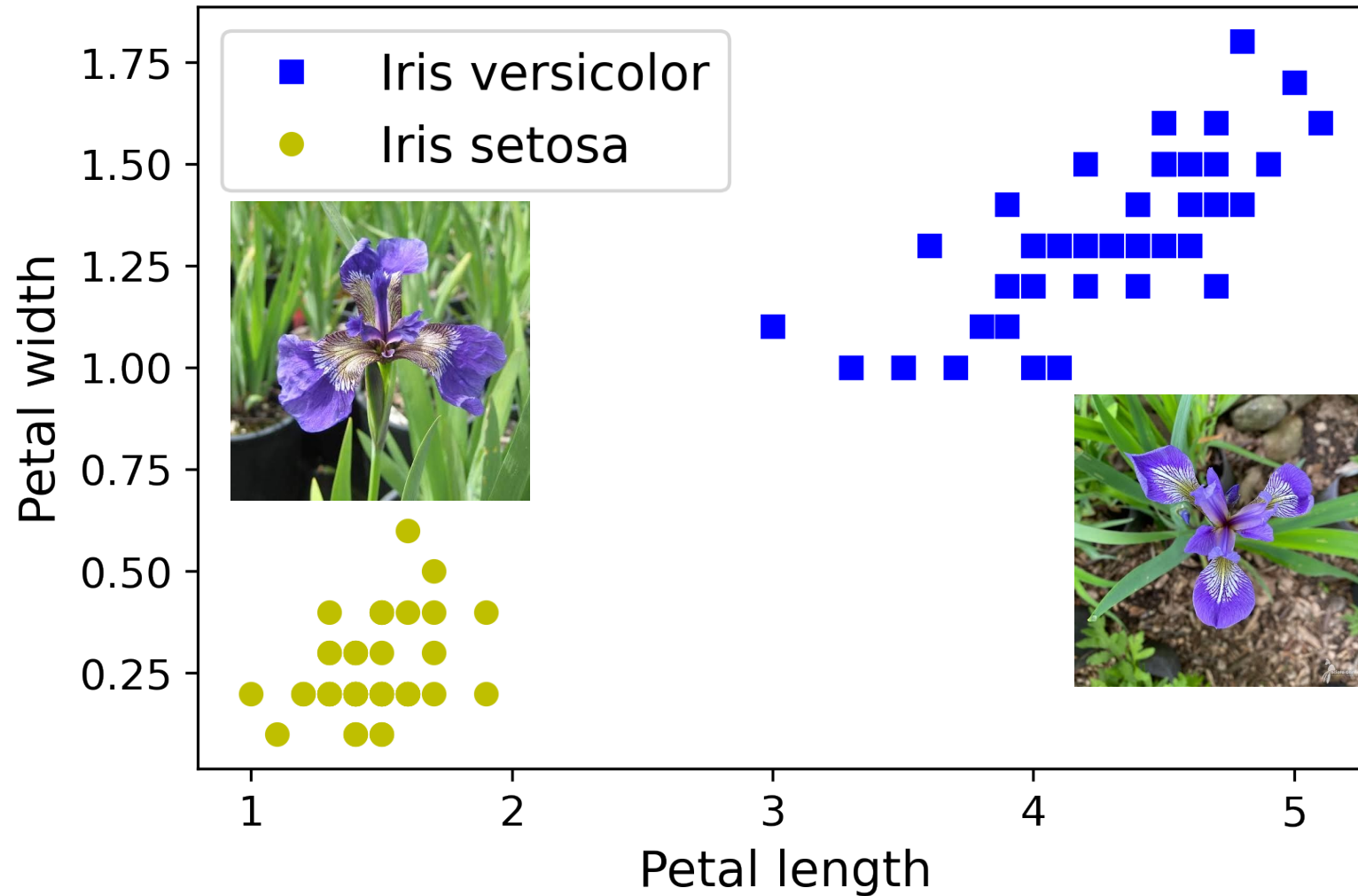
<https://colab.research.google.com/drive/1N4g0QDmCV9r0Hg6eLD7fdcRXXHJLSfM?usp=sharing>

Support Vector Machine (SVM)

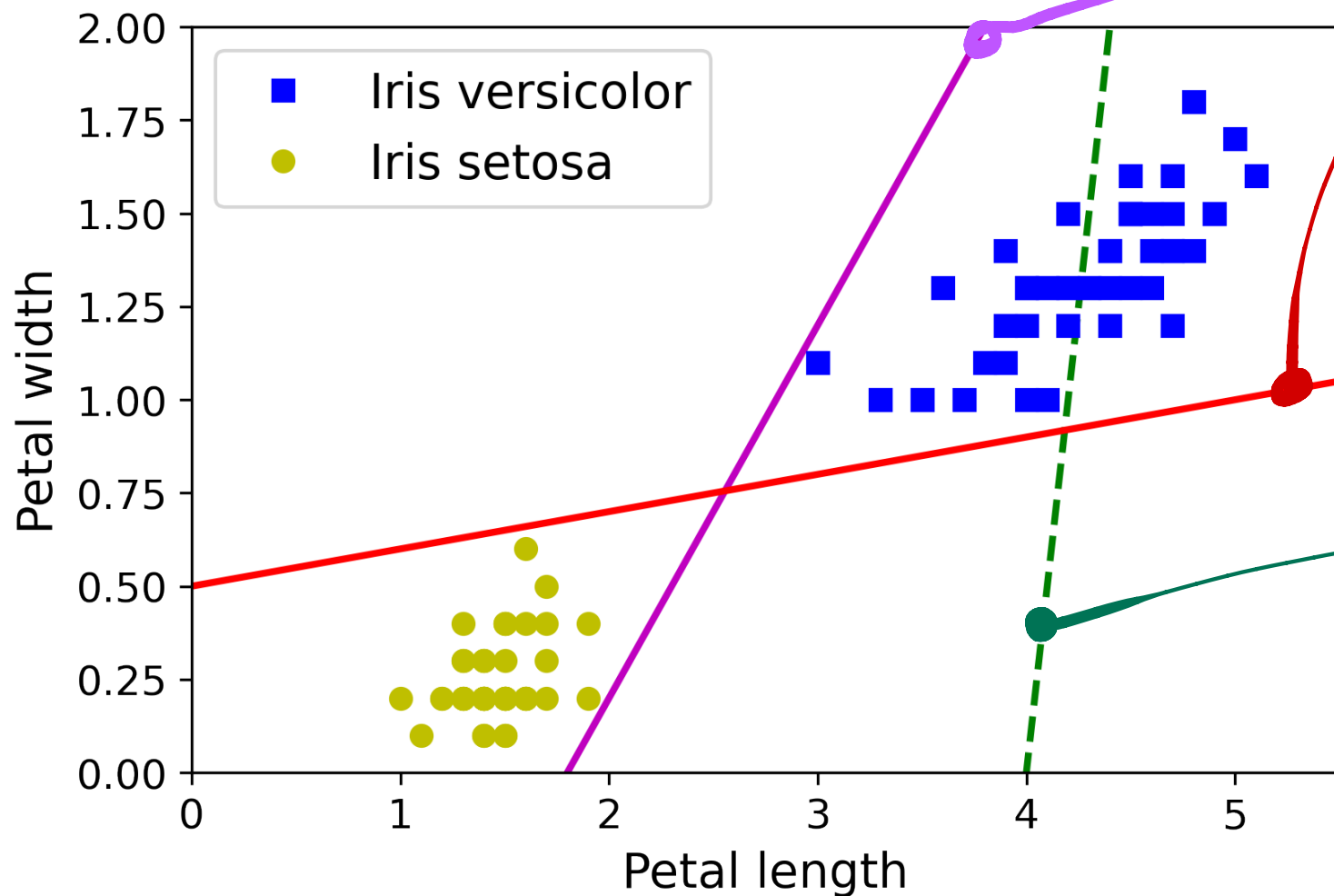
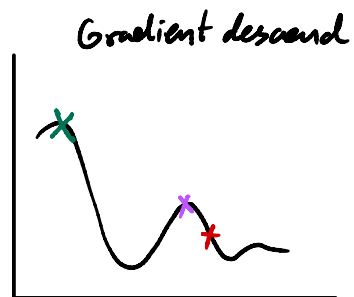
- A Support Vector Machine (SVM) is a very powerful and versatile Machine Learning model, capable of performing linear or nonlinear classification, regression, and even outlier detection.
- SVMs are particularly well suited for classification of complex but small-or medium-sized datasets.



Linearly separable



Which line?



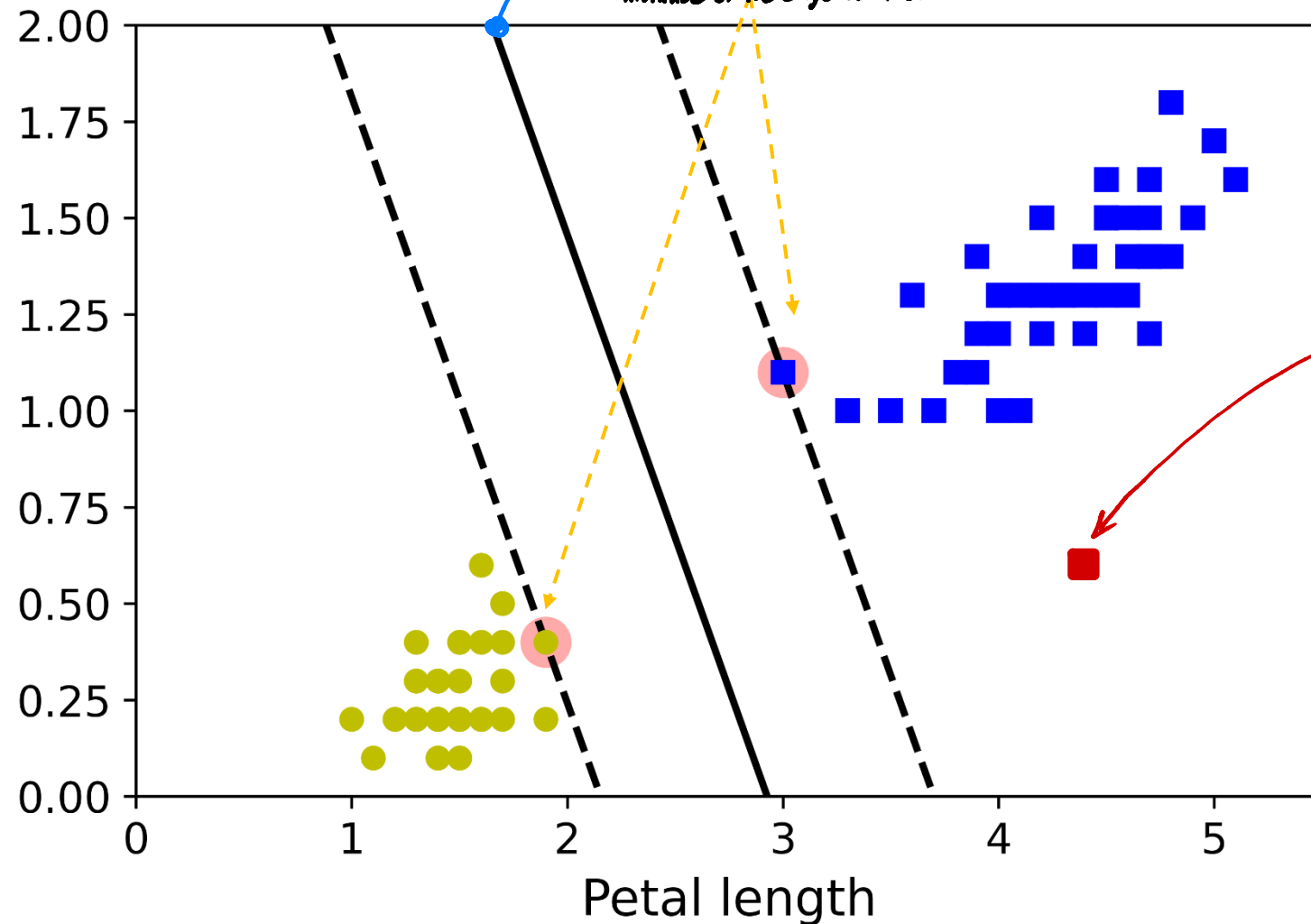
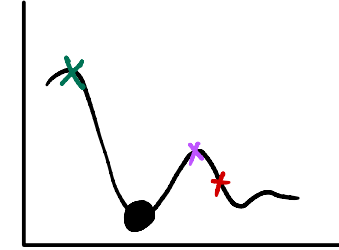
ไม่ได้แค่แบ่ง class ที่อยู่ห่างจาก training instance ที่ใกล้ที่สุดด้วย
 จินตนาการว่ามีเส้นประ: ตีเส้นไว้ให้เงิน ถนนที่กว้างที่สุด

Support vectors

Decision Boundary
 ของ SVM Classifier

Support vectors
 instance of the edge of street

Gradient descent



ถ้ามี instance ใน
 อยู่นอกถนน
 จะไม่มาพาด: ไรกับ
 decision boundary

∴ It is fully determined
 (supported) by support vectors

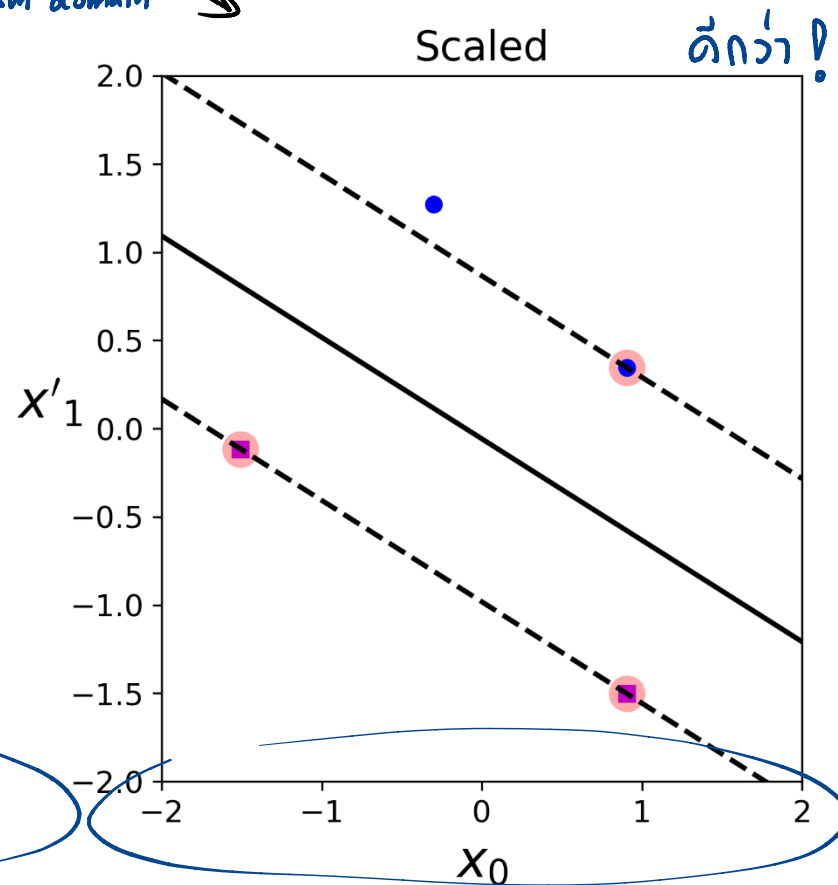
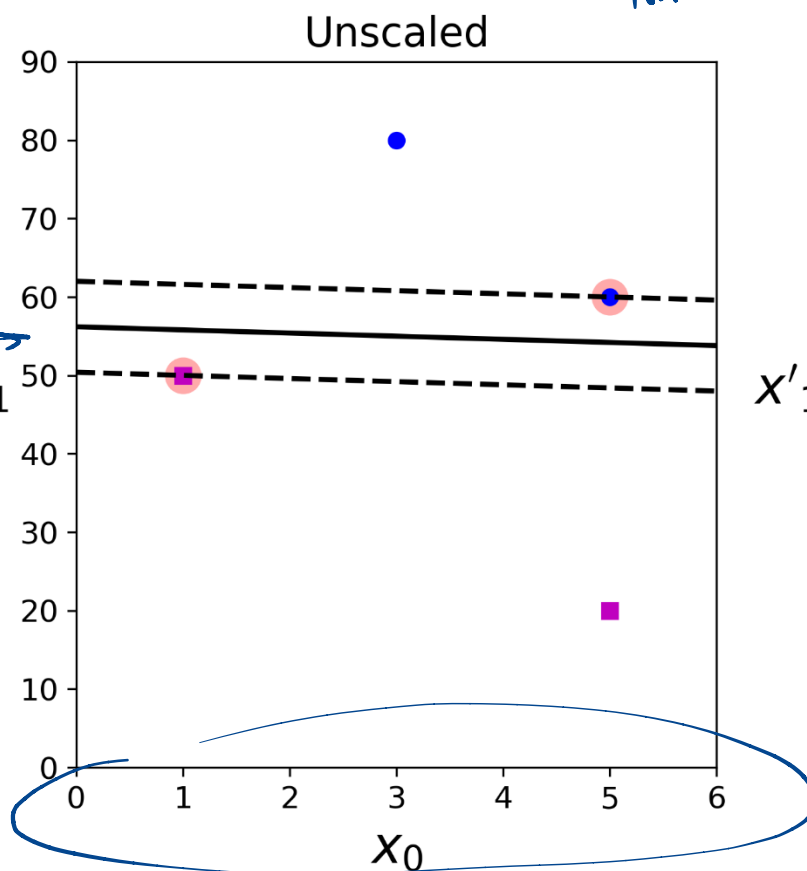
SVMs are sensitive to feature scales

Feature scaling

Normalization
/ Standardization

Transform domain

กรณี mis-classify
ส่วนนี้



ดีขึ้น!

Scale ให้เท่ากัน

Decision function: linear SVM

Inference →

$$\hat{y} = \begin{cases} 0 & \text{if } \mathbf{w}^T \mathbf{x} + b < 0, \\ 1 & \text{if } \mathbf{w}^T \mathbf{x} + b \geq 0 \end{cases}$$

The linear SVM classifier model predicts the class of a new instance $\mathbf{x}$ by simply computing the decision function $\mathbf{w}^T \mathbf{x} + b = w_1 x_1 + \dots + w_n x_n + b$: if the result is positive, the predicted class $\hat{y}$ is the positive class (1), or else it is the negative class (0); see

$\mathbf{w}$ = weight vector (controls slope/tilt of the boundary)

b = bias (shifts the boundary up/down or left/right)

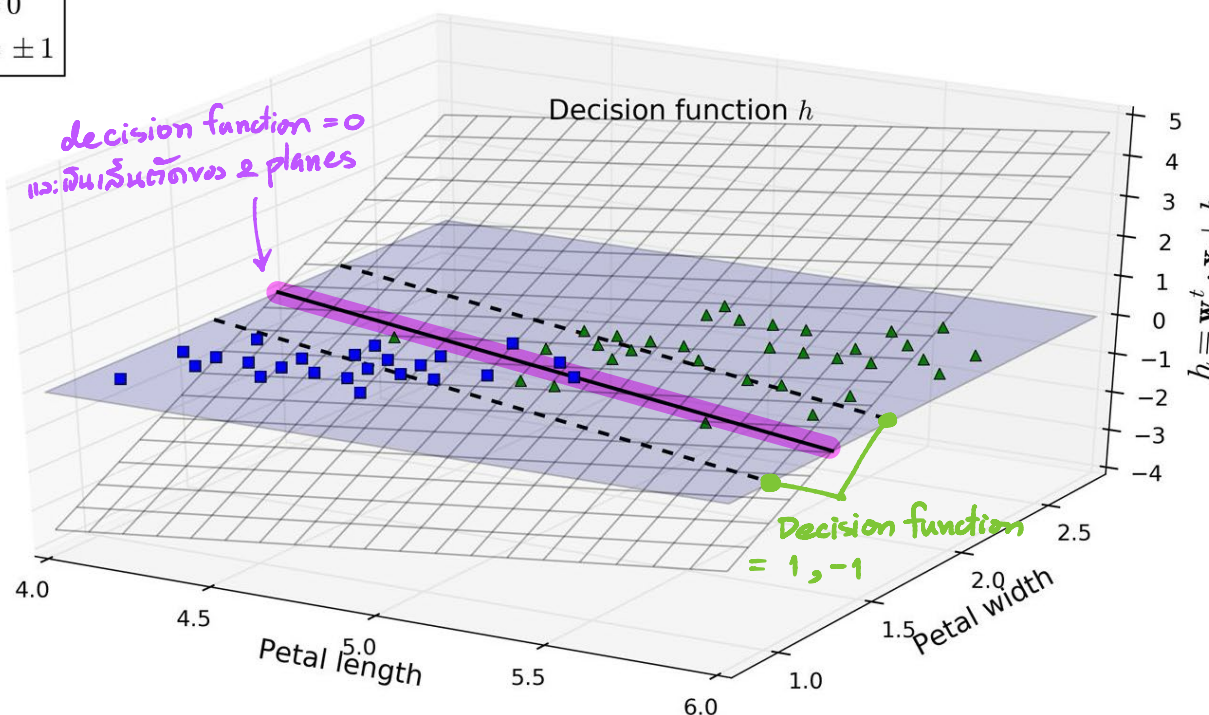
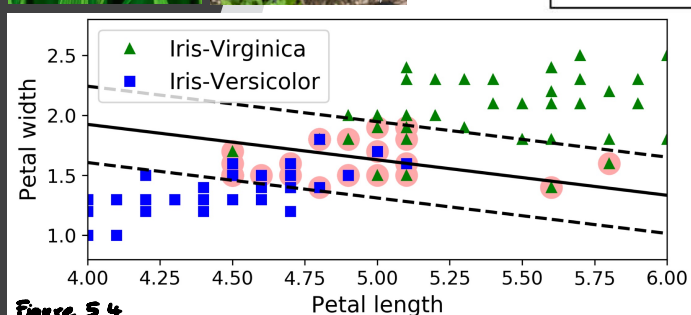
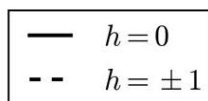
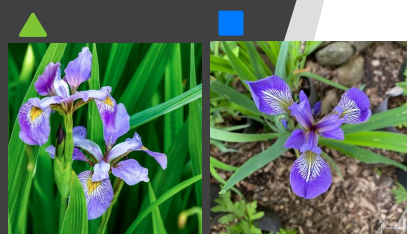


Figure 5-12 shows the decision function that corresponds to the model on the left of Figure 5-4: it is a two-dimensional plane since this dataset has two features (petal width and petal length). The **decision boundary** is the set of points where the decision function is equal to 0: it is the **intersection of two planes**, which is a straight line (represented by the thick solid line).

The dashed lines represent the points where the decision function is equal to 1 or -1: they are parallel and at equal distance to the decision boundary, forming a margin around it.

Training a linear SVM classifier means finding the value of $\mathbf{w}$ and $\mathbf{b}$ that make this margin as wide as possible while avoiding margin violations (hard margin) or limiting them (soft margin).

Figure 5-12. Decision function for the iris dataset

Training objective

$\div 2$
 Slope $w_1 = 1 = \frac{\Delta y}{\Delta x} = \frac{1}{1}$
 Slope $w_1 = 0.5 = \frac{\Delta y}{\Delta x} = \frac{1}{2}$ $\times (\text{margin})$
 เพิ่มเป็น 2 เท่า
 $\therefore$ เรา weigh vector (slope) ที่น้อย
 margin (Δx) ก็เพิ่มมากขึ้นด้วย

- The smaller the weight vector, the larger the margin

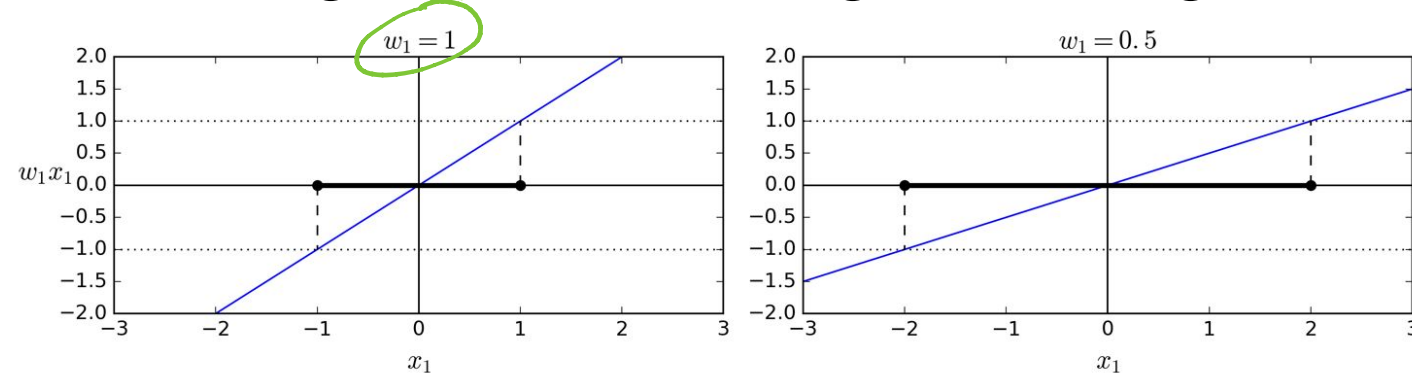


Figure 5-13. A smaller weight vector results in a larger margin

- Margin violation. Let $t^{(i)} = -1$ for negative instance (if $y^{(i)} = 0$) and $t^{(i)} = 1$ for positive instance (if $y^{(i)} = 1$)

Ex: $w = -4, b = 3$

Negative Instance: $t_1 = -1, x_1 = 1$

Positive Instance: $t_2 = 1, x_2 = 3$

$$(-1)(-4(1) + 3) \geq 1$$

$$1 \geq 1$$

Classify OK

$$(1)(-4(3) + 3) \geq 1$$

$$-1 \neq 1$$

Classify ผิด ตัวอย่าง Neg instance

$$t^{(i)}(\mathbf{w}^T \mathbf{x}^{(i)} + b) \geq 1$$

กำหนดให้เป็นจริง แปลว่า Classify OK

target (label มา train) $\therefore$ SVM เป็น supervised

Hard margin linear classifier

Objective function

$$\underset{\mathbf{w}, b}{\text{minimize}} \quad \frac{1}{2} \mathbf{w}^T \mathbf{w}$$

$$\text{subject to} \quad t^{(i)}(\mathbf{w}^T \mathbf{x}^{(i)} + b) \geq 1 \quad \text{for } i = 1, 2, \dots, m$$

Constraint

$\mathbf{w} \uparrow \quad \mathbf{x} \text{ កំណត់ } + b \text{ កំណត់}$

$\|\mathbf{w}\|^2$

Minimize $\|\mathbf{w}\|$ ទារ $\because$ diff ក៏ $\mathbf{w} = 0$ មិន
 លទ្ធផលឡើយ $\frac{1}{2} \|\mathbf{w}\|^2 \Rightarrow \frac{1}{2} \mathbf{w}^T \mathbf{w}$ ៖ diff ងាយ
 $\because$ វាជាការបំប្លែង

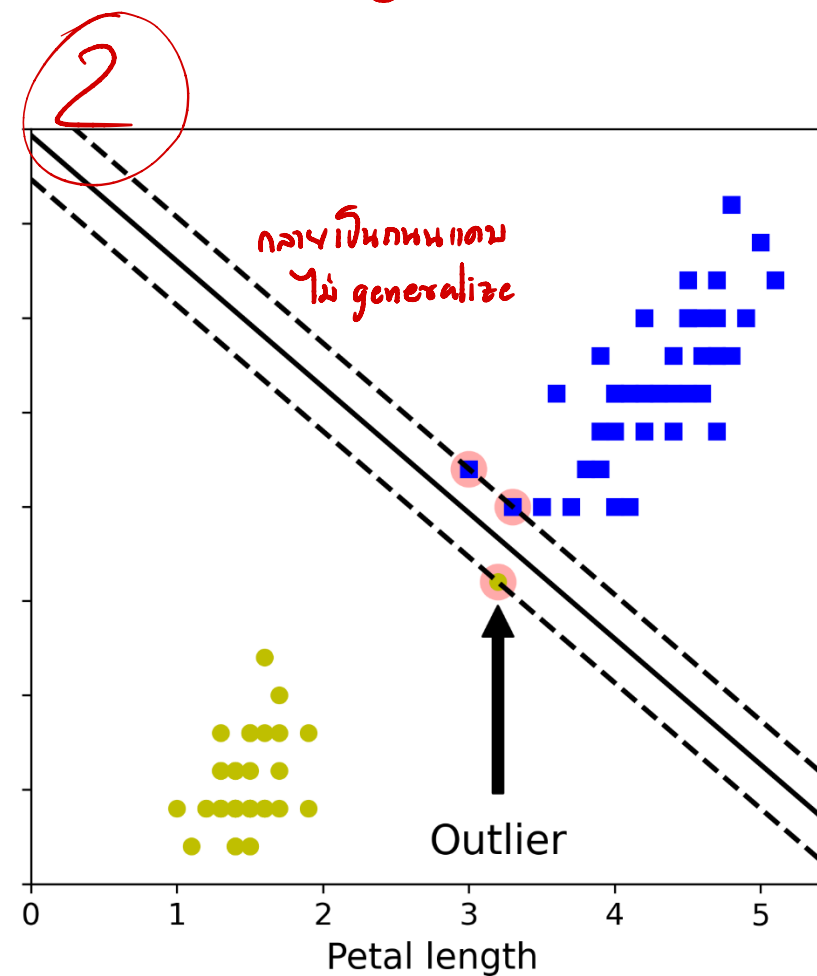
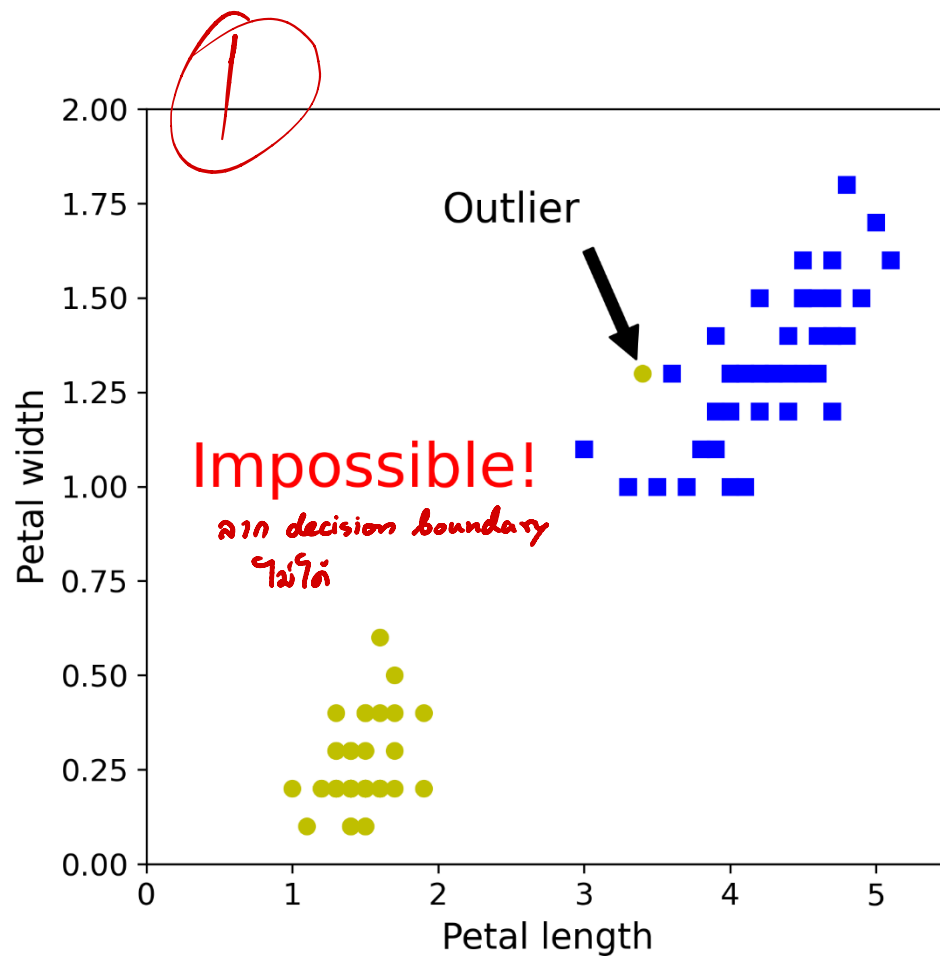
Sensitivity to outliers

Hard margin Classification – all instances

ตัวอย่างของกรณี และปัญหาคืออะไร

ข้อ 2 ข้อ 1 ① ใช้กับ data ที่ไม่ใช่ class แบบ linear

② Sensitive กับ outliers

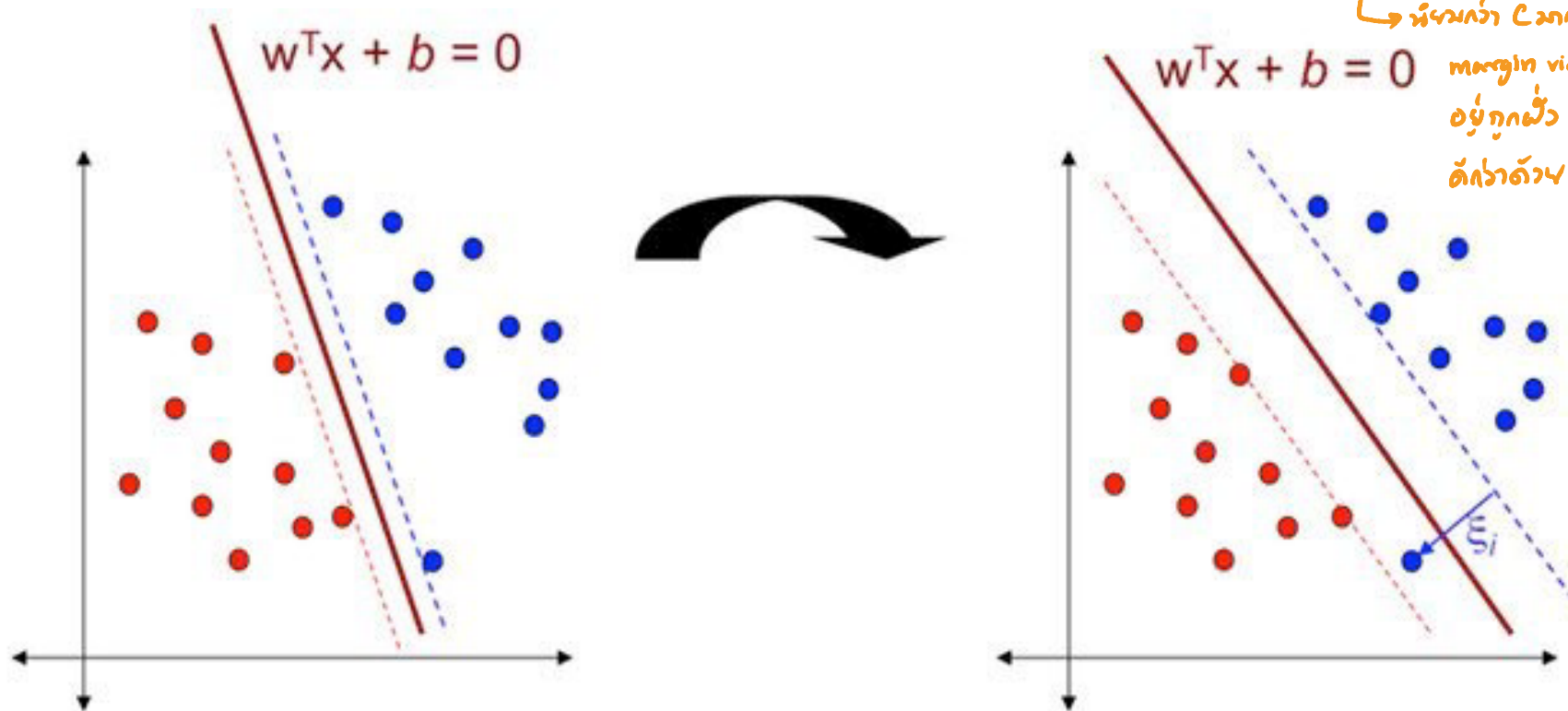


Soft margin classification

เพื่อแก้ปัญหาของ Hard margin classification
 ที่ต้องการ balance ระหว่าง ทำให้ margin กว้างที่สุดเท่าที่จะเป็นไปได้
 กับ ข้อผิดพลาดของ margin violation
 instance ที่อยู่ใกล้ขอบ หรือ อยู่ผิดฝั่ง

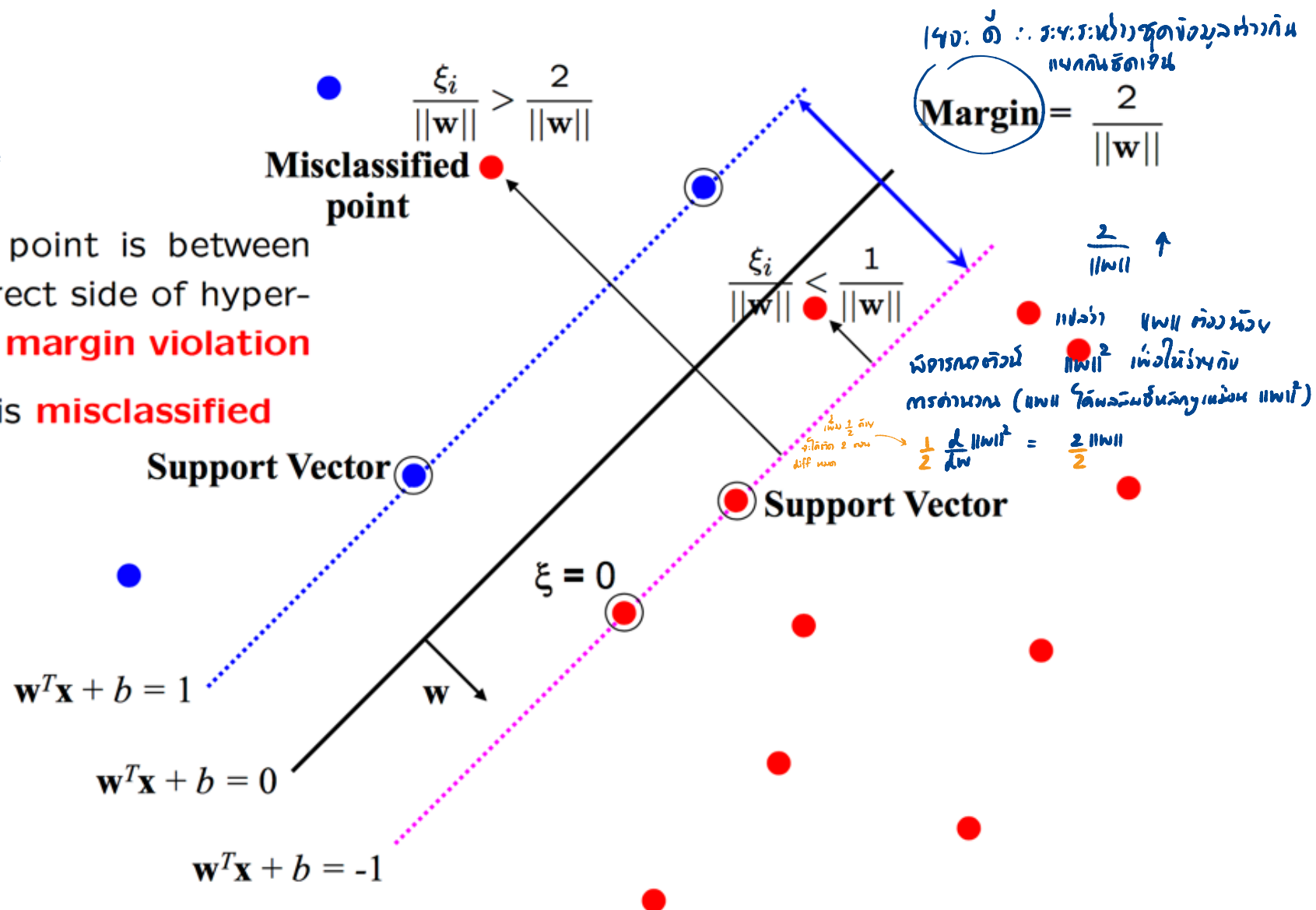
Note: ใน Scikit-Learn's SVM มี C hyperparameter
 C ใหญ่ — ขอบ margin แคบ และ margin violation

→ หมายความว่า C ใหญ่ :: เมื่อ C ใหญ่จะ
 margin violation: ให้ความสำคัญกับ
 อยู่ถูกฝั่ง แทนที่จะ generalize
 ได้กว้าง (ลด overfitting)



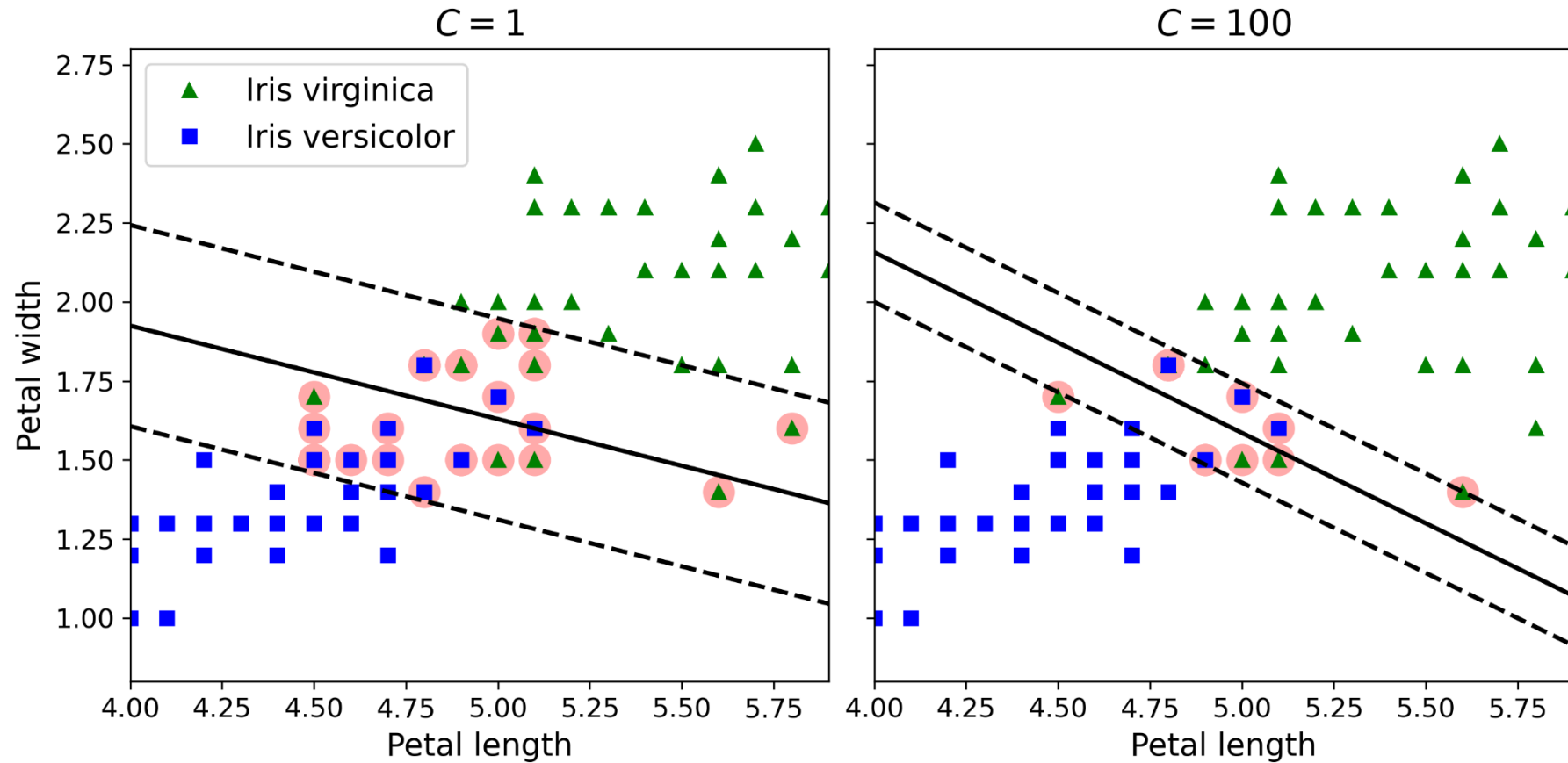
Slack Variable $\xi_i \geq 0$

- for $0 < \xi \leq 1$ point is between margin and correct side of hyperplane. This is a **margin violation**
- for $\xi > 1$ point is **misclassified**



Regularization via margin

Note: In Scikit-Learn's SVM C hyperparameter
 C คือ — ความกว้างของ margin
 → C มีค่าสูง :: margin น้อย
 margin น้อย :: โมเดลไม่ generalize
 โมเดลไม่ generalize :: overfitting (ลด accuracy)



Soft margin linear classifier

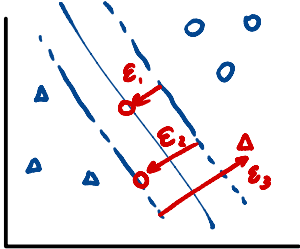
Objective function

$$\underset{\mathbf{w}, b, \zeta}{\text{minimize}} \quad \frac{1}{2} \mathbf{w}^T \mathbf{w} + C \sum_{i=1}^m \zeta^{(i)}$$

subject to $t^{(i)}(\mathbf{w}^T \mathbf{x}^{(i)} + b) \geq 1 - \zeta^{(i)}$ and $\zeta^{(i)} \geq 0$ for $i = 1, 2, \dots, m$

Constraints

Annotations:
 - $C \sum_{i=1}^m \zeta^{(i)}$: violation of constraints (red text)
 - $\zeta^{(i)}$: margin Error (purple text)
 - $C \sum_{i=1}^m \zeta^{(i)}$: constraints are hard margin (purple text)



- C is a regularization parameter:
 - small C allows constraints to be easily ignored $\rightarrow$ large margin
 - large C makes constraints hard to ignore $\rightarrow$ narrow margin
 - $C = \infty$ enforces all constraints: hard margin

Solving hard and soft linear SVM Quadratic programming problem

Equation 5-5. Quadratic Programming problem

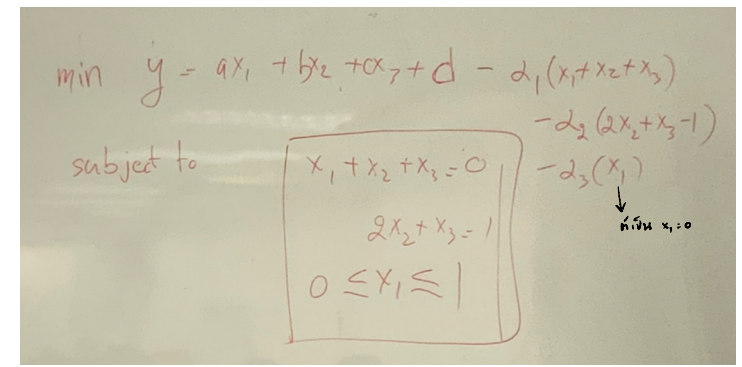
$$\begin{array}{ll} \text{Minimize}_{\mathbf{p}} & \frac{1}{2} \mathbf{p}^T \mathbf{H} \mathbf{p} + \mathbf{f}^T \mathbf{p} \\ \text{subject to} & \mathbf{A} \mathbf{p} \leq \mathbf{b} \end{array}$$

where

$$\left\{ \begin{array}{l} \mathbf{p} \text{ is an } n_p \text{-dimensional vector } (n_p = \text{number of parameters}), \\ \mathbf{H} \text{ is an } n_p \times n_p \text{ matrix,} \\ \mathbf{f} \text{ is an } n_p \text{-dimensional vector,} \\ \mathbf{A} \text{ is an } n_c \times n_p \text{ matrix } (n_c = \text{number of constraints}), \\ \mathbf{b} \text{ is an } n_c \text{-dimensional vector.} \end{array} \right.$$

Dual problem

Lagrange multiplier



min $y = ax_1 + bx_2 + cx_3 + d$
 subject to $x_1 + x_2 + x_3 = 0$
 $2x_2 + x_3 = 1$
 $0 \leq x_1 \leq 1$

$- \alpha_1(x_1 + x_2 + x_3)$
 $- \alpha_2(2x_2 + x_3 - 1)$
 $- \alpha_3(x_1)$
 $\downarrow$
 $\alpha_1 = 0$

- Transform constrained optimization objective into an unconstrained one

- For example,

$$\text{minimize } f(x,y) = x^2 + 2y$$

$$\text{subjective } 3x + 2y + 1 = 0 \quad (\text{equality constrain})$$

- Lagrangian

$$\text{minimize } g(x,y,\alpha) = f(x,y) - \alpha(3x + 2y + 1)$$

α is Lagrange

ကဲ $(\hat{x}, \hat{y})$ ပဲသ်သ်တလဝသ် constrained optimization problem

သ်သ်တလဝသ် $\hat{\alpha}$ ကဲကဲကဲ $(\hat{x}, \hat{y}, \hat{\alpha})$ ပဲသ် stationary point ဝသ် Lagrangian

(ဒုတိယ derivative ကဲကဲ 0)

သ်သ် partial derivative $g(x, y, \alpha)$ ဝသ် x, y, α ကဲကဲ partial derivative = 0

$$\text{ဒုတိယသ်သ်သ်သ်} \left\{ \begin{array}{l} \frac{\partial}{\partial x} g(x, y, \alpha) = 2x - 3\alpha \\ \frac{\partial}{\partial y} g(x, y, \alpha) = 2 - 2\alpha \\ \frac{\partial}{\partial \alpha} g(x, y, \alpha) = -3x - 2y - 1 \end{array} \right\} = 0$$

သ်သ် partial derivatives ပဲသ် 0

$$2\hat{x} - 3\hat{\alpha} = 2 - 2\hat{\alpha} = -3\hat{x} - 2\hat{y} - 1 = 0$$

$$\text{သ်သ်သ်} \quad \hat{x} = \frac{3}{2}, \quad \hat{y} = -\frac{11}{4}, \quad \hat{\alpha} = 1$$

Dual problem Generalized Lagrangian

$$\begin{aligned} & \underset{\mathbf{w}, b}{\text{minimize}} && \frac{1}{2} \mathbf{w}^T \mathbf{w} \\ & \text{subject to} && t^{(i)}(\mathbf{w}^T \mathbf{x}^{(i)} + b) \geq 1 \quad \text{for } i = 1, 2, \dots, m \end{aligned}$$

Generalized Lagrangian vs hard margin

$\alpha^{(i)}$ as Karush-Kuhn-Tucker (KKT) multiplier

ใช้ Lagrange
คูณกับ constraint

Equation C-1. Generalized Lagrangian for the hard margin problem

$$\mathcal{L}(\mathbf{w}, b, \alpha) = \frac{1}{2} \mathbf{w}^T \mathbf{w} - \sum_{i=1}^m \alpha^{(i)} (t^{(i)}(\mathbf{w}^T \mathbf{x}^{(i)} + b) - 1)$$

with $\alpha^{(i)} \geq 0$ for $i = 1, 2, \dots, m$

Partial derivative of generalized Lagrangian

Equation C-2. Partial derivatives of the generalized Lagrangian

$$\textcircled{1} \quad \nabla_{\mathbf{w}} \mathcal{L}(\mathbf{w}, b, \alpha) = \overset{\frac{\partial}{\partial \mathbf{w}} \frac{1}{2} \mathbf{w}^T \mathbf{w}}{= \frac{\partial}{\partial \mathbf{w}} \frac{1}{2} \mathbf{w}^2} \mathbf{w} - \sum_{i=1}^m \alpha^{(i)} t^{(i)} \mathbf{x}^{(i)}$$

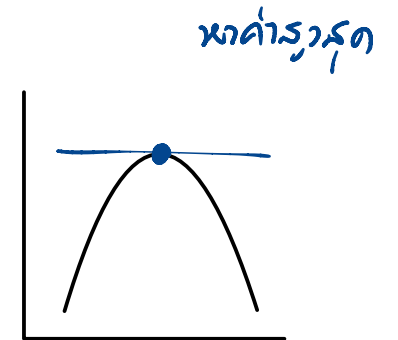
$$\textcircled{2} \quad \frac{\partial}{\partial b} \mathcal{L}(\mathbf{w}, b, \alpha) = - \sum_{i=1}^m \alpha^{(i)} t^{(i)}$$

Stationary points, set to extrema

Equation C-3. Properties of the stationary points

$$\textcircled{1} \quad W - \sum_{i=1}^m \hat{\alpha}^{(i)} t^{(i)} x^{(i)} = 0 \rightarrow \hat{\mathbf{W}} = \sum_{i=1}^m \hat{\alpha}^{(i)} t^{(i)} \mathbf{x}^{(i)}$$

$$\textcircled{2} \quad - \sum_{i=1}^m \hat{\alpha}^{(i)} t^{(i)} = 0 \rightarrow \sum_{i=1}^m \hat{\alpha}^{(i)} t^{(i)} = 0$$



Substitute to generalized Lagrangian Loss function

$$\mathcal{L}(w, b, \alpha) = \frac{1}{2} w^T w - \sum_{i=1}^m \alpha^{(i)} (t^{(i)} (w^T x^{(i)} + b) - 1)$$

$$= \frac{1}{2} w^T w - \sum_{i=1}^m \alpha^{(i)} t^{(i)} x^{(i)T} w - \sum_{i=1}^m \alpha^{(i)} t^{(i)} b + \sum_{i=1}^m \alpha^{(i)}$$

$$= \frac{1}{2} w^T w - w^T w + \sum_{i=1}^m \alpha^{(i)}$$

$$w = \sum_{i=1}^m \alpha^{(i)} t^{(i)} x^{(i)} : \quad -\frac{1}{2} w^T w + \sum_{i=1}^m \alpha^{(i)}$$

Equation C-4. Dual form of the SVM problem

$$\mathcal{L}(\hat{w}, \hat{b}, \alpha) = \frac{1}{2} \sum_{i=1}^m \sum_{j=1}^m \alpha^{(i)} \alpha^{(j)} t^{(i)} t^{(j)} \mathbf{x}^{(i)T} \mathbf{x}^{(j)} - \sum_{i=1}^m \alpha^{(i)}$$

with $\alpha^{(i)} \geq 0$ for $i = 1, 2, \dots, m$

Dual problem

Objective function

* รวมกัน

Equation 5-6. Dual form of the linear SVM objective

$$\underset{\alpha}{\text{minimize}} \frac{1}{2} \sum_{i=1}^m \sum_{j=1}^m \alpha^{(i)} \alpha^{(j)} t^{(i)} t^{(j)} \mathbf{x}^{(i)T} \mathbf{x}^{(j)} - \sum_{i=1}^m \alpha^{(i)}$$

subject to $\alpha^{(i)} \geq 0$ for $i = 1, 2, \dots, m$

Primal solution

* สอนมสสท

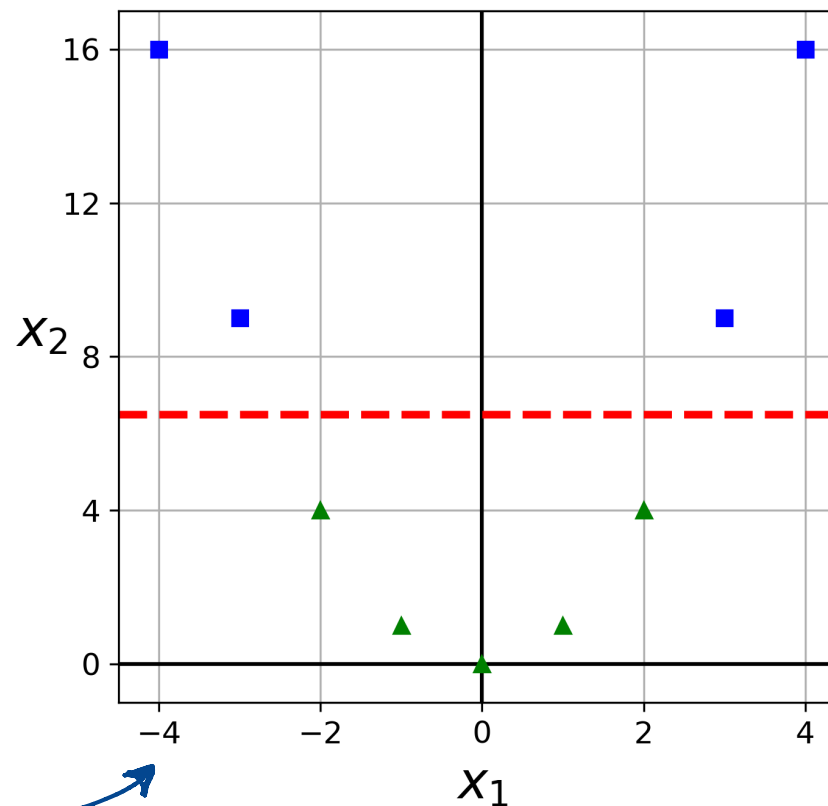
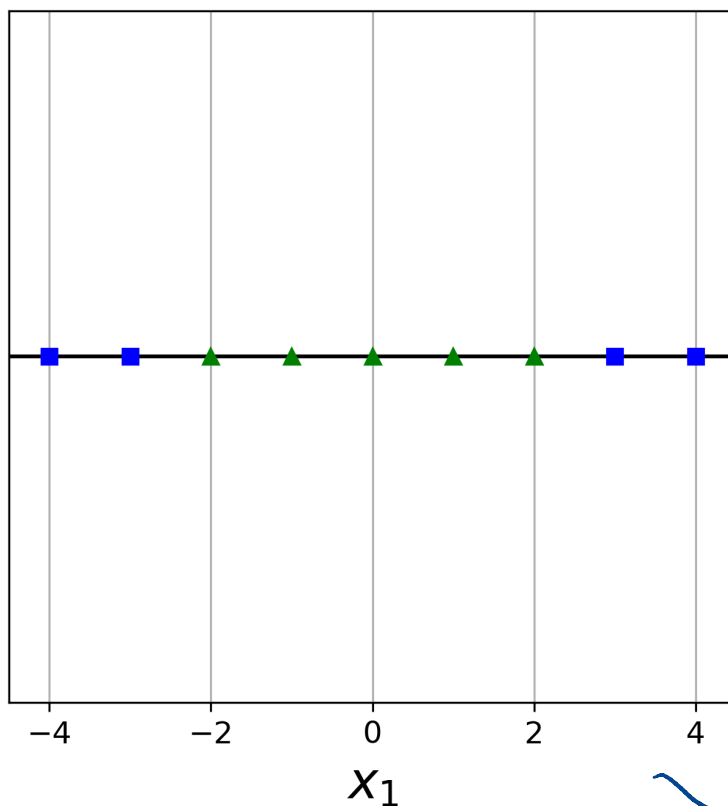
- After obtaining the solution of dual problem, $\alpha^{(i)}$, we can use them to compute the solution of primal problem as follows

$$\hat{\mathbf{w}} = \sum_{i=1}^m \hat{\alpha}^{(i)} t^{(i)} \mathbf{x}^{(i)}$$

$$\hat{b} = \frac{1}{n_s} \sum_{\substack{i=1 \\ \hat{\alpha}^{(i)} > 0}}^m (t^{(i)} - \hat{\mathbf{w}}^T \mathbf{x}^{(i)})$$

Nonlinear SVM

Adding features to make a dataset linearly separable



$$x_2 = (x_1)^2$$

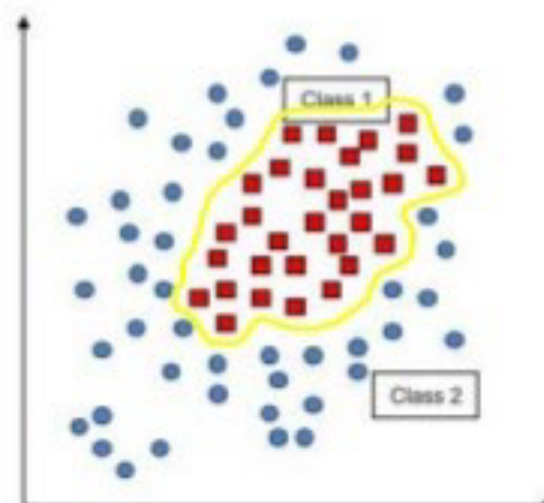
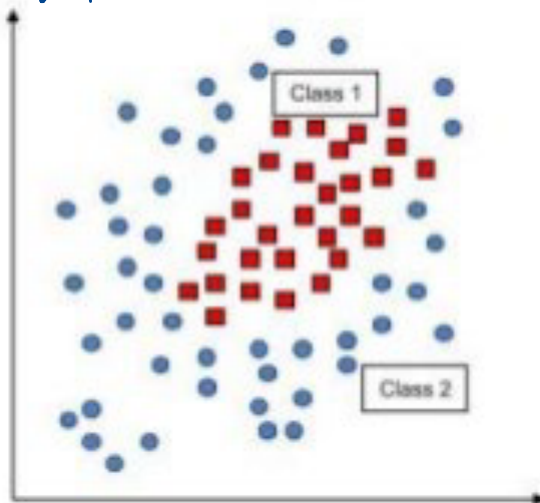
เพิ่ม feature ให้มันตัวเส้นแบ่งได้

การเพิ่ม Polynomial Features คือการสร้างฟีเจอร์ใหม่จากฟีเจอร์เก่า $\rightarrow$ ยิ่ง degree สูง ยิ่งเยอะและซับซ้อน

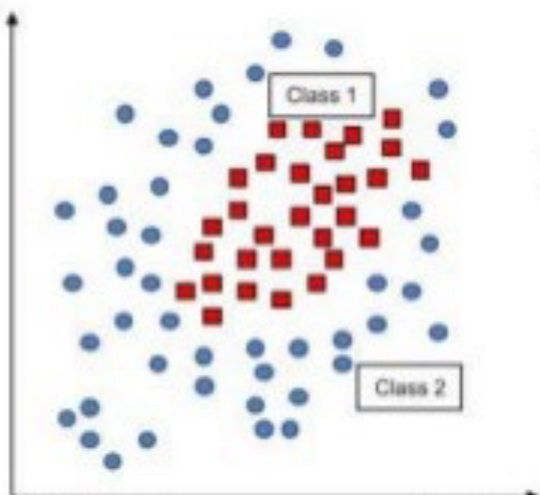
Polynomial Kernel ใน SVM ใช้ Kernel Trick $\rightarrow$ ได้ผลเหมือนเพิ่มฟีเจอร์มหาศาล แต่ไม่ต้องสร้างจริง $\rightarrow$ เร็วกว่า ประหยัดกว่า และยังรองรับข้อมูลที่ซับซ้อนได้ดี

SVM Kernel

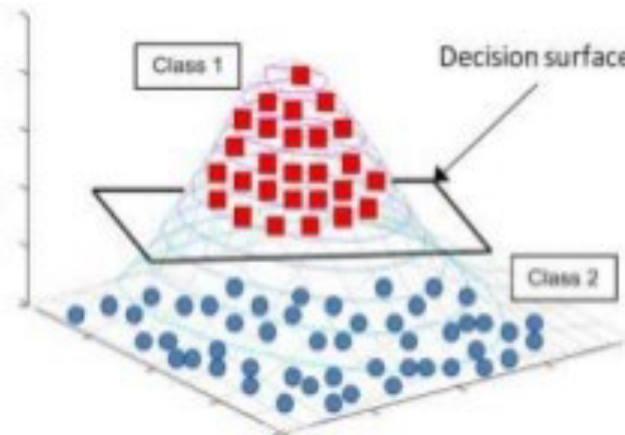
ก. เพิ่ม feature เรียง



Non Linear
Decision
Boundary



kernel



Kernel
method

ใช้ clustering ไม่ได้ $\because$ clustering ดู distance



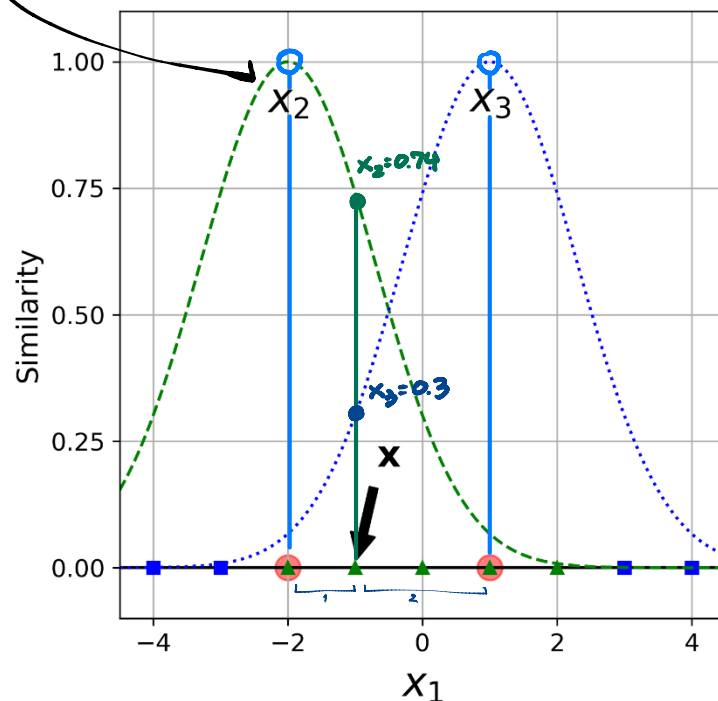
เท่านั้นที่คิดกันมาก

Adding similarity feature

Equation 5-1. Gaussian RBF

$$\phi_{\gamma}(\mathbf{x}, \ell) = \exp(-\gamma \|\mathbf{x} - \ell\|^2)$$

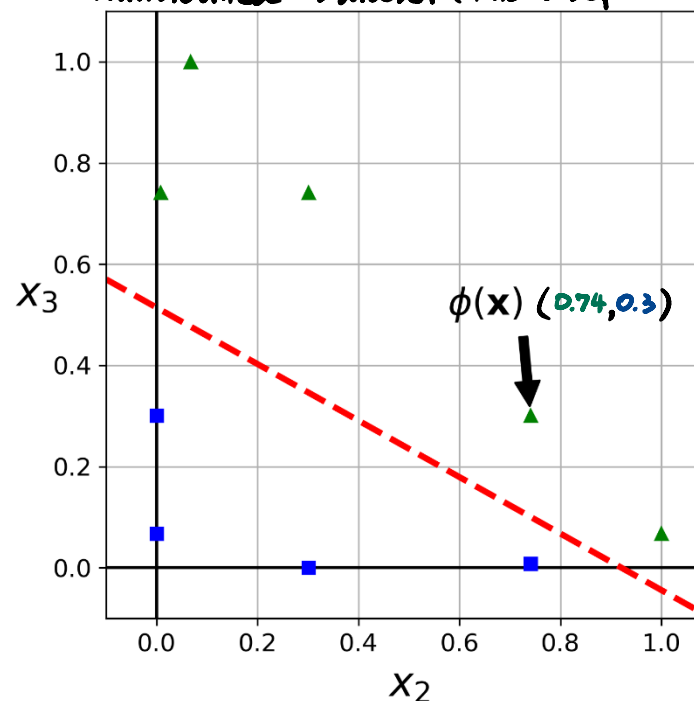
รูปร่าง bell-shaped
ค่าอยู่ระหว่าง [0, 1]
→ symmetric landmark



Spread

Mean

Transformed Dataset (linear separable)



∴ อยากรู้ 2Dimension (ง่าย combination ที่สุด)
STEP1 หนัก 2 landmark (ง่ายสุด combination → อยากรู้ linearly separable)
หนัก 2 landmark $x_1 = -2$ $y_1 = 1$
หนัก 2 landmark $x_1 = 1$ $y_1 = 2$
instance $x_1 = -1$
STEP2 หา feature ใหม่ (Let $\gamma = 0.3$)
 $x_2 = \exp(-0.3 \times 1^2) \approx 0.74$
 $x_3 = \exp(-0.3 \times 2^2) \approx 0.3$

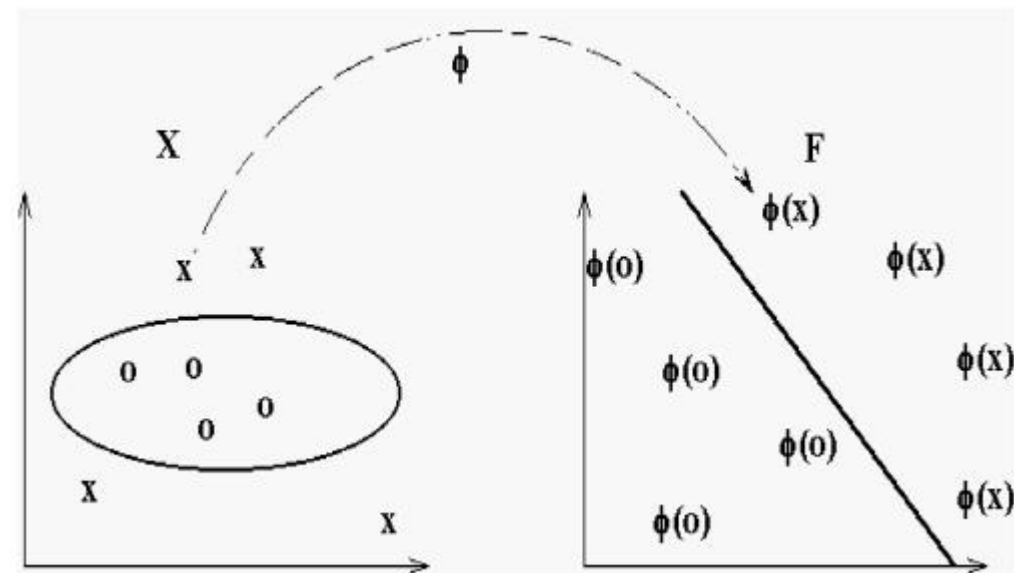
Kernelized SVM

Second-degree polynomial

సంకలనం

Equation 5-8. Second-degree polynomial mapping

$$\phi(\mathbf{x}) = \phi\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right) = \begin{pmatrix} x_1^2 \\ \sqrt{2} x_1 x_2 \\ x_2^2 \end{pmatrix}$$



Kernel trick

Equation 5-9. Kernel trick for a 2nd-degree polynomial mapping

$$\begin{aligned}\phi(\mathbf{a})^T \phi(\mathbf{b}) &= \begin{pmatrix} a_1^2 \\ \sqrt{2} a_1 a_2 \\ a_2^2 \end{pmatrix}^T \begin{pmatrix} b_1^2 \\ \sqrt{2} b_1 b_2 \\ b_2^2 \end{pmatrix} = a_1^2 b_1^2 + 2a_1 b_1 a_2 b_2 + a_2^2 b_2^2 \\ &= (a_1 b_1 + a_2 b_2)^2 = \left(\begin{pmatrix} a_1 \\ a_2 \end{pmatrix}^T \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} \right)^2 = (\mathbf{a}^T \mathbf{b})^2\end{aligned}$$

Recall the objective function



Equation 5-6. Dual form of the linear SVM objective

$$\begin{aligned} \underset{\alpha}{\text{minimize}} & \frac{1}{2} \sum_{i=1}^m \sum_{j=1}^m \alpha^{(i)} \alpha^{(j)} t^{(i)} t^{(j)} \underbrace{\mathbf{x}^{(i)T} \mathbf{x}^{(j)}}_{\text{All pairs}} - \sum_{i=1}^m \alpha^{(i)} \\ \text{subject to} & \alpha^{(i)} \geq 0 \quad \text{for } i = 1, 2, \dots, m \end{aligned}$$

Replacing with $\phi(\mathbf{x}^{(i)})^T \phi(\mathbf{x}^{(j)})$

For 2-nd degree polynomial kernel, $\phi(\mathbf{x}^{(i)})^T \phi(\mathbf{x}^{(j)}) = (\mathbf{x}^{(i)T} \mathbf{x}^{(j)})^2$

Common kernel

Equation 5-10. Common kernels

Linear: $K(\mathbf{a}, \mathbf{b}) = \mathbf{a}^T \mathbf{b}$

Polynomial: $K(\mathbf{a}, \mathbf{b}) = (\gamma \mathbf{a}^T \mathbf{b} + r)^d$

Gaussian RBF: $K(\mathbf{a}, \mathbf{b}) = \exp (-\gamma \| \mathbf{a} - \mathbf{b} \|^2)$

Sigmoid: $K(\mathbf{a}, \mathbf{b}) = \tanh (\gamma \mathbf{a}^T \mathbf{b} + r)$

Prediction with kernelized SVM

Equation 5-11. Making predictions with a kernelized SVM

$$\begin{aligned}
 h_{\hat{\mathbf{w}}, \hat{b}}(\phi(\mathbf{x}^{(n)})) &= \hat{\mathbf{w}}^T \phi(\mathbf{x}^{(n)}) + \hat{b} = \left(\sum_{i=1}^m \hat{\alpha}^{(i)} t^{(i)} \phi(\mathbf{x}^{(i)}) \right)^T \phi(\mathbf{x}^{(n)}) + \hat{b} \\
 &= \sum_{i=1}^m \hat{\alpha}^{(i)} t^{(i)} (\phi(\mathbf{x}^{(i)})^T \phi(\mathbf{x}^{(n)})) + \hat{b} \\
 &= \sum_{\substack{i=1 \\ \hat{\alpha}^{(i)} > 0}}^m \hat{\alpha}^{(i)} t^{(i)} K(\mathbf{x}^{(i)}, \mathbf{x}^{(n)}) + \hat{b}
 \end{aligned}$$

w វ៉ិចទ័រ (vector)
 មិនមាន w ក៏គណនា scalar
 ល្អជាង → គិតលឿន

No need of w!!

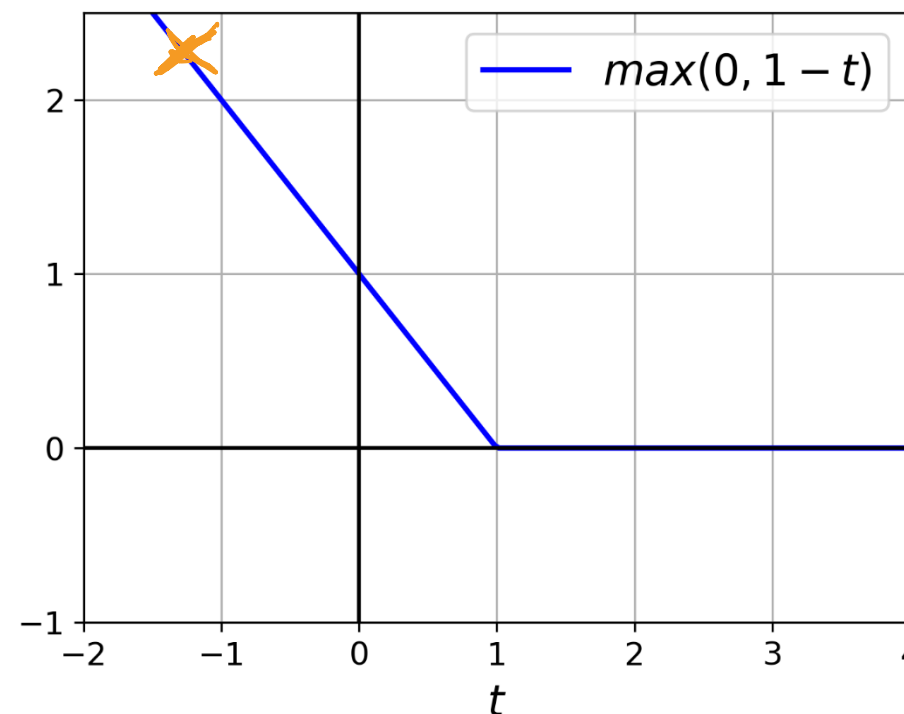
Bias term for kernel trick

Equation 5-12. Computing the bias term using the kernel trick

$$\begin{aligned}\hat{b} &= \frac{1}{n_s} \sum_{\substack{i=1 \\ \hat{\alpha}^{(i)} > 0}}^m (t^{(i)} - \hat{\mathbf{w}}^T \phi(\mathbf{x}^{(i)})) = \frac{1}{n_s} \sum_{\substack{i=1 \\ \hat{\alpha}^{(i)} > 0}}^m \left(t^{(i)} - \left(\sum_{j=1}^m \hat{\alpha}^{(j)} t^{(j)} \phi(\mathbf{x}^{(j)}) \right)^T \phi(\mathbf{x}^{(i)}) \right) \\ &= \frac{1}{n_s} \sum_{\substack{i=1 \\ \hat{\alpha}^{(i)} > 0}}^m \left(t^{(i)} - \sum_{\substack{j=1 \\ \hat{\alpha}^{(j)} > 0}}^m \hat{\alpha}^{(j)} t^{(j)} K(\mathbf{x}^{(i)}, \mathbf{x}^{(j)}) \right)\end{aligned}$$

Hinge Loss

- Derivative equals to -1 when $t^{(i)} < 1$
- This makes it suitable for using with Gradient Descent in online learning เรีนโด้ว ทักวณโด้ว ค่องย ปรังโด้



$$J(\mathbf{w}, b) = \frac{1}{2} \mathbf{w}^T \mathbf{w} + C \sum_{i=1}^m \max(0, 1 - t^{(i)}(\mathbf{w}^T \mathbf{x}^{(i)} + b))$$

$\max(0, 1 - ((1)(-2)(3) + 3))$
 $\max(0, 4)$
 4

Ex: ✍

ผิด! กำลัง Error



End of Lecture 3

Question?

